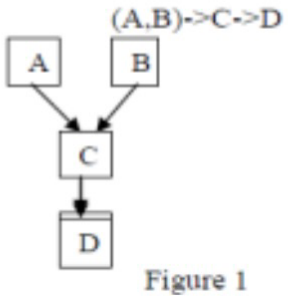
Unit -1	
1	Explain features of java with Example.
2	Explain JVM, JDK and JRE
3	What is variable? Give one example program for defining variables in java. Also list rules for valid variable names.
4	Explain this keyword with Example.
5	Explain static keyword with Example.
6	Explain finalize() and garbage collection with Example.
7	Explain Ragged Array With Examples.
8	Write a java program that creates and initialises a four integer element array. Calculate and display the average of its values.
9	Declare a class called employee having employee_id andemployee_name as members. Extend class employees to have a subclass called salary having designation and monthly_salary as members. Define following: 1. Required constructors 2. A method to find and display all details of employees drawing salaries more than Rs. 20000/-. 3. Method main for creating an array for storing these details given as command line arguments and showing usage of above methods.
10	Differentiate String class and StringBuffer class with explanation of its methods.

11	Differentiate StringBuilder class and StringBuffer class with explanation of its methods
12	What is a Wrapper class in Java? Explain with examples.
13	There is no destructor in Java. Justify.
14	Differentiate POP and OOP.

15	State whether any error exists in the following code. If so, correct the error and give output.
	class Test {
	public static void main(String args[]) {
	A a = new A();
	a.print();
	}
	}
	class A {
	String s;
	A(String s) {
	this.s=s;
	}
	public void print() {
	System.out.println(s);
	}
	}

16	Write a java program to do the sum of command line arguments passed two Double numbers.
17	Discuss public, private, protected and default access modifiers with examples. OR What is a visibility modifier? Explain all with examples.
18	Explain recursion in java with example.

Unit -2	
1	Differentiate abstract class and interface with example.
2	Explain inheritance and its types with Example.
3	Why multiple inheritance is not supported in java.
4	What is package? Explain type of packages.
5	Explain ways to import packages with examples.
6	Explain method overriding with example.
7	Differentiate Method Overloading and Method Overriding with examples.
8	Explain super keywords.
9	Explain Final keywords.
10	Explain Dynamic method dispatch or runtime polymorphism.
11	Explain instanceof operator.
12	Explain single level and multiple inheritances in java. Write a program to demonstrate a combination of both types of inheritance as shown in figure.  Figure 1
13	What is error? Explain various types of errors. How can we handle run time errors in java?
14	Explain the importance of Exception handling in java. Write a program to handle ArithmeticException, ArrayIndexOutofBoundsException using try-catch-finally and throw.

15	What is Is-A relationship in Java?
16	Are constructor and instance initialization block inherited to subclass?
17	Are static members inherited to subclass in Java?

18	Will the code successfully compiled? If yes, what is the output?
	public class A {
	  int x = 20;
	}
	public class B extends A {
	  int x = 30;
	}
	public class Test {
	public static void main(String[] args)
	{
	  B b = new B();
	  System.out.println(b.x);
	
	  A a = new A();
	  System.out.println(a.x);
	
	  A a2 = new B();
	  System.out.println(a2.x);
	  }
	}

19	Will this code run successfully? If yes, what will be the output?
	public class A
	{
	  private int x = 50;
	  void m1() {
	  System.out.println(x);
	  }
	}
	public class B extends A {
	
	}
	public class Test {
	public static void main(String[] args)
	{
	  A a = new B();
	  a.m1();
	  }
	}

20	Can I import same package/class twice? Will the JVM load the package twice at runtime?

Unit -3	
1	Explain File constructors, any two methods of class File and method seek.
2	Explain use of Array List collection class with example.
3	Explain use of Linked List collection class with example.
4	Explain use of Vector collection class with example.
5	Explain use of Enumeration with examples.
6	Explain Client-Server Communication in java networking with proper diagrams only [No Code].
7	Write a client-server program using TCP sockets to echo the message sent by the client.
8	Write the use of InetAddress class. List and explain the methods of it.
9	Explain java.util package with examples.
10	Differentiate Byte Streams and Character Streams in java with suitable example programs for both.
11	Explain FileOutputStream in java with an example program.
12	Explain FileInputStream in java with an example program.
13	Explain the following terms with respect to exception handling. i) try ii) catch iii) throw iv) throws v) finally
14	Explain Built in Exception.
15	Explain Custom Exception
16	Explain Throwable Class.
17	Give the difference between throw and throws keywords with example.
18	Write an application that generates custom exception if any value from its command line arguments is negative.
19	Write the program to print the IP address of the local machine.

20	What is Server Socket? How does it work in java ? Explain it with an example.
21	Explain the following classes with their use. 1. DatagramSocket 2. DatagramPacket class
22	List and write the use of classes available in java.net Package.

Unit -4	
1	Draw and explain the life cycle of thread. Also list and explain various methods of thread.
2	What is multitasking? And how to achieve it?
3	What is Multithreading and Multiprocessing with examples? Write a program to create two threads with thread class and runnable interface.
4	What do you mean by Generic Method in Java? Explain Generic class constructors with an example program.
5	What is Generics? Explain wildcards with examples.
6	Explain Applet initialization and termination with example.
7	What is priority in thread? Explain Thread Priority with an example.
8	Explain applet life cycle with proper diagram.
9	Write a program to create a thread by implementing Runnable Interface.
10	Explain: wait, sleep with example. OR Compare the methods of wait and sleep.

5	What is Generics? Explain wildcards with examples.
6	Explain Applet initialization and termination with example.
7	What is priority in thread? Explain Thread Priority with an example.
8	Explain applet life cycle with proper diagram.
9	Write a program to create a thread by implementing Runnable Interface.
10	Explain: wait, sleep with example. OR Compare the methods of wait and sleep.
11	Explain synchronisation in thread with example.
12	Explain the problem occurs in thread without synchronisation.
13	Explain Thread communication.
14	Explain Deadlock.
15	Differentiate Thread class and Runnable interface.
16	What is the difference between an Applet and a Java Application ?
17	What are the ways to resize an Applet?
18	What is the role of the destroy() method in an Applet?
19	What are signed Applets and unsigned Applets?
20	How can you pass parameters to Applets?